Block 5 Technical Report
Switch Network

Sampling Transmission Gates of a 1.8-V Switched-Capacitor Delta-Sigma Modulator

SKY130 transistor-level characterization with Xschem and ngspice

Closure status. Block 5 was closed by the project owner on 2 November 2026 for the scope *nominal, pre-layout, simulation* (TT, 1.8 V, 27 °C). All results are *measured, not final*: there is no PVT, mismatch, noise or post-layout evidence, and no acceptance limit was imposed on the hold error.

Revision 1 — 2 November 2026

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1 Executive Summary

Block 5 is the sampling network of the second-order switched-capacitor delta-sigma modulator: low-threshold (LVT) CMOS transmission gates that sample the signal and the feedback reference onto capacitors. This report characterizes the gate as a track-and-hold (experiments E1–E8), verifies the MIM-capacitor model used in the netlists, and records the transistor-level closed-loop verification of the whole modulator that closed the block.

- **Hold error.** Turning the switch off leaves a signal-dependent pedestal of +3.302 mV (signal 1.1 V) and −1.834 mV (signal 0.7 V) on a 1 pF hold capacitor, mirrored between the two sides. Input steps during the hold phase produced no measurable feedthrough or leakage (below 0.05 mV).
- **Scaling.** Doubling the hold capacitor scales the pedestal by 0.52 (P) and 0.50 (N); a 5-ns gate edge gives 0.76 and 1.02.
- **Aperture.** The switch changes from tracking an input step to ignoring it within about 3 ns (from 3 ns before the gate fall to its start); steps arriving ≥ 1 ns after the fall begins are ignored.
- **Linearity.** Across 0.2 V to 1.6 V the pedestal varies from −3.19 mV to +9.28 mV. Differentially this is a gain error of +0.964% and a peak deviation from a straight line of 1.36 mV (0.0485% of the 2.8 V span).
- **Closed loop.** The transistor-level modulator reaches 75.37 dB SNDR at OSR 80 (4096-point FFT, −6 dBFS, noiseless transient simulation), against 77.16 dB for the behavioral model, with second-order noise shaping.
- **Specification.** A second-order 1-bit modulator cannot reach the original 99-dB target at OSR 80. The project owner revised the target to ≈75 dB at OSR 80 (25 kHz bandwidth).

Status convention used throughout: **measured, not final** (a simulation result at the nominal point, not a qualification) and **simulated, not final pass** (a functional run that behaved as expected without a numeric acceptance limit). The word "pass" is never used for a qualification.

2 Block Role and Test Conditions

The block uses SKY130 low-threshold devices: PMOS $W/L = 8/0.35$ µm and NMOS $W/L = 4/0.15$ µm in each transmission gate. The standalone track-and-hold testbench (`Block5_SH_LVT`) places a 1 pF hold capacitor on each side, a 100 Ω source resistance, and a common mode of 0.9 V. The sampling clock `PHI_SH` has a 250 ns period (4 MHz); it is high from 31 to 141 ns (track) and falls between 141 and 142 ns. Differential signals are ±0.2 V around the common mode unless stated. All runs use ngspice-46, the SKY130A TT corner, 1.8 V and 27 °C, with testbenches built in Xschem 3.4.8RC.

Each experiment changes one variable relative to the baseline E1, except E6, which changes the device type and, with it, the PMOS length. Pedestal is defined as $v_{hold}(t_{off} + 13\,\text{ns}) - v_{hold}(t_{off} - 2\,\text{ns})$ for E1–E6 and as $v_{hold}(t_0 + 155) - v_{hold}(t_0 + 139)$ for E8.

3 Hold Error (E1–E6)

3.1 Baseline

In E1 the input changes 18 ns after the switch opens in both cycles. Figure 1 shows the hold voltages and the turn-off transient.

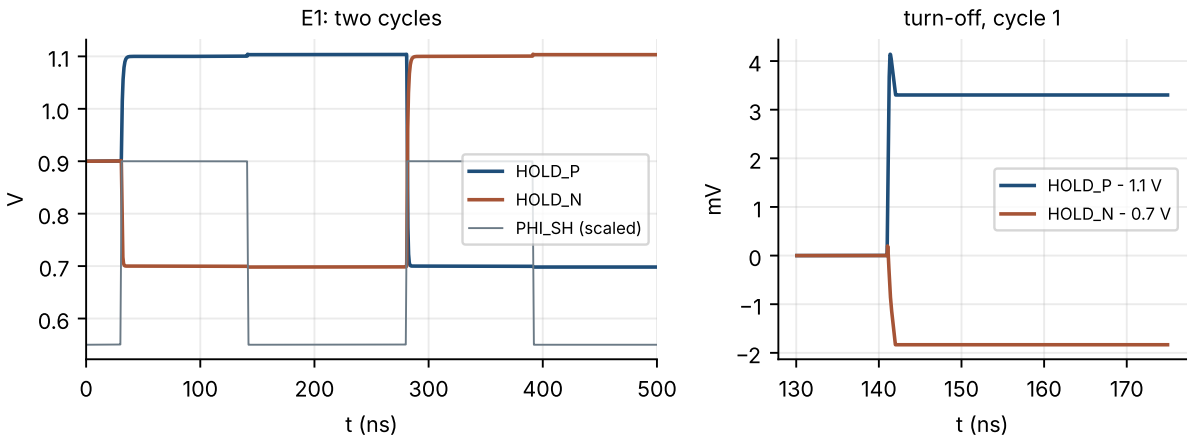


Figure 1: E1 baseline. Left: hold voltages over two clock cycles. Right: turn-off transient of the first cycle, referred to the tracked level.

3.2 Sensitivity

Table 1: Pedestal of cycle 1 (P: 1.1 V, N: 0.7 V) and ratio to E1.

Exp.	Change	P (mV)	N (mV)	P/E1	N/E1
E1	baseline (input steps 18 ns after turn-off)	+3.302	−1.833	1.00	1.00
E2	input frozen during hold	+3.302	−1.833	1.00	1.00
E3	input step 98 ns after turn-off	+3.302	−1.833	1.00	1.00
E4	hold capacitor 2 pF	+1.713	−0.914	0.52	0.50
E5	gate edges 5 ns	+2.522	−1.872	0.76	1.02
E6	standard-V_t gate (PMOS $L = 0.15\,\mu$m)	+0.512	−1.441	0.16	0.79

Observations. (i) E2 and E3 are indistinguishable from E1: the pedestal is created at turn-off and the input source does not couple into the held node afterwards; the immediate and drifting responses to the input step were below 0.05 mV in E1 and E3. (ii) Doubling the hold capacitor halves the pedestal, as expected for a charge-injection mechanism ($\Delta V \propto 1/C_H$). (iii) The pedestal of the opposite sides is mirrored: the same level produces the same pedestal on P and N. (iv) E6 differs in two ways at once (device threshold and PMOS length), so it is a comparison between two designs and not a single-variable result.

Earlier smoke test. A dynamic smoke test recorded before these experiments showed an apparent hold change of +24.95 mV (P) and −14.21 mV (N) at 450 ns. The reconstructed baseline E1 does not reproduce it (its stimulus is a reconstruction; the original was not available). The aperture experiment below offers a consistent explanation, which has not been proven.

4 Aperture Window (E7)

In E7 the input step is moved, cycle by cycle, from $\delta = -20$ to +12 ns relative to the start of the gate fall.

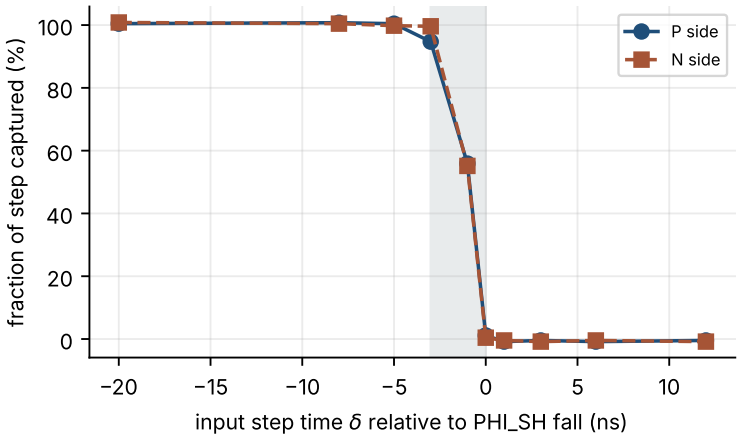


Figure 2: Fraction of the input step captured on the hold capacitor versus the step time. The shaded band marks the transition.

Table 2: Captured fraction of the input step (hold value 30 ns after the fall, relative to the previous level).

δ (ns)	P side	N side
≤ -5	100.4 – 100.8%	99.8 – 100.8%
-3	94.7%	99.6%
-1	55.9%	55.1%
0	1.1%	0.4%
$\geq +1$	pedestal only	pedestal only

A step that arrives 3 ns or more before the gate fall is tracked completely; a step arriving at or after the start of the fall is almost entirely rejected; the transition occupies about 3 ns. In the earlier smoke test the apparent change at 450 ns corresponds to a captured fraction of 6.2% (P) and 3.6% (N), which lies inside this transition. This is consistent with the input stepping inside the aperture, but it is a hypothesis, because the original stimulus timing is unknown. *Design consequence:* in the modulator the sampled input must be stable for about 3 ns before the sampling switch opens; the non-overlapping clock generator (Block 6) has to guarantee this margin.

5 Pedestal versus Signal Level (E8)

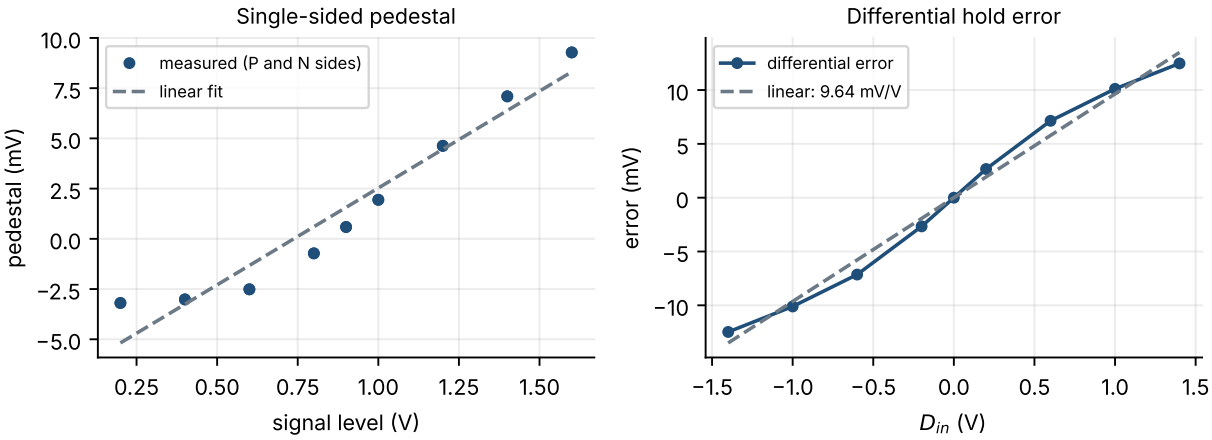


Figure 3: E8. Left: single-sided pedestal versus signal level. Right: differential hold error versus differential input.

Table 3: Single-sided pedestal versus level. P and N sides agree to 0.002 mV at equal level.

Level (V)	0.2	0.4	0.6	0.8	0.9	1.0	1.2	1.4	1.6
Pedestal (mV)	−3.19	−3.02	−2.51	−0.72	+0.59	+1.94	+4.63	+7.09	+9.28

The pedestal is nearly flat at low levels (about 1 mV/V between 0.2 and 0.4 V) and steep from 0.8 V upward (about 11 mV/V to 13 mV/V), the signature of an NMOS/PMOS charge cancellation that moves with the signal level. Differentially, the error follows $0.000 + 9.640\,\text{mV/V} \cdot D_{in}$: a gain error of +0.964% and a peak residual of 1.36 mV to the straight line (0.0485% of the 2.8 V span, −66.3 dB). A cubic fit $aD + bD^3$ to the four positive points gives $b = -1.68\,\text{mV/V}^3$ (compressive; fit residual ≤ 0.34 mV) and a rough third-harmonic estimate of −88 dB, −76 dB and −68 dB for differential amplitudes of 0.3 V, 0.6 V and 1.0 V. This estimate is derived from four points and is not a sine-wave simulation.

6 MIM Capacitor Model Check

A dedicated AC test compared the Xschem MIM symbol with an ideal 1 pF capacitor (the method recovered 1000.00 fF). The symbol writes both `MF=N` and `m=N`; the model ignores `MF`, and only `m` multiplies. One unit capacitor is 2.645 fF: the symbol with `MF=100` measures 264.50 fF and with `MF=1000` 2645.0 fF. Raw variants confirm that a value of `MF=100` with `m=1` gives only one unit. No double multiplicity occurs.

7 Closed-Loop Verification of the Modulator

7.1 Setup

The netlist contains both integrator stages, the StrongARM comparator, two B01 buffers, the SR latch, the 1-bit DAC ($V_{REFP} = 1.2$ V, $V_{REFN} = 0.6$ V, mid-level 0.9 V) and the feedback sampling

branches, at $f_s = 4\,\text{MHz}$. Two findings from the first smoke runs matter for any reader reproducing it: the stages are non-inverting, and the DAC branch of *both* stages must use the swapped connection; the straight connection in stage 2 is positive feedback and drives the second integrator to the rails.

7.2 Results

DC test. For a differential input of 100 mV the mean DAC level was +0.1002 V against the expected 0.1 V, with no rail excursion.

Sine test. Input: 300 mV differential peak (−6 dBFS for a 0.6 V reference span), coherent bin 13 of a 4096-point record ($f_{in} = 12.70\,\text{kHz}$), 5 µs reset, 16 warm-up cycles, Hann window, signal bins 13 ± 3, in-band bins 3–25.

Table 4: Closed-loop sine test, OSR 80.

	Transistor level	Behavioral model
SNDR	75.37 dB	77.16 dB
ENOB	12.23	–
Model SNDR over 16 input phases	–	75.3 to 78.8 dB
Mean of the 4096 analysed bits	0.0000	–
Output common mode, stage 1 / stage 2	0.859–0.916 / 0.860–0.926 V	–

The transistor-level SNDR lies inside the spread of the model, so the two cannot be distinguished at this record length. The result is insensitive to the analysis settings (74.6 to 75.7 dB for signal half-widths of 2 to 4 bins and 1 to 5 skipped low bins).

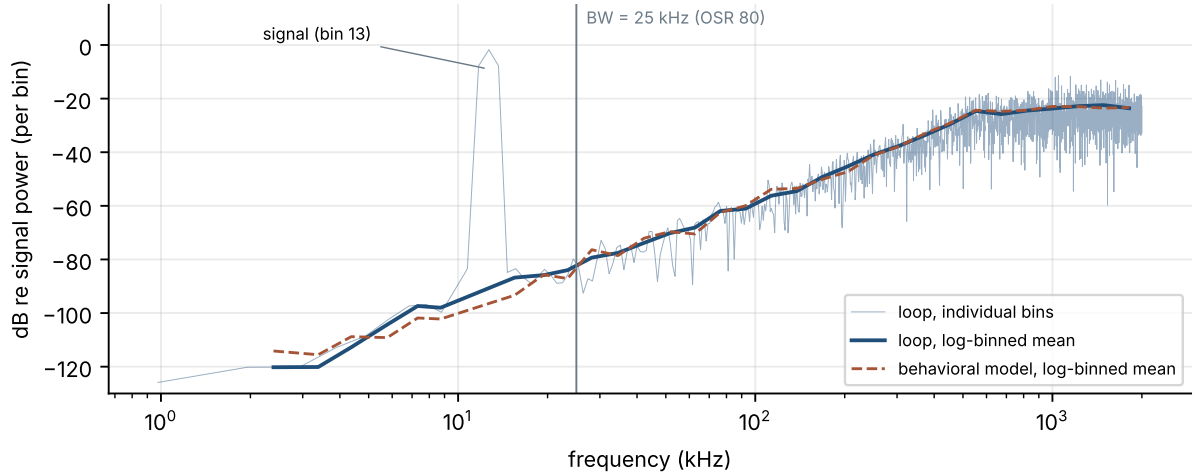


Figure 4: Spectrum of the 4096-bit output stream of the transistor-level modulator and of the behavioral model.

Table 5: Noise power per frequency band (dB relative to the signal power of each run; signal bins excluded).

Bins	Transistor level	Model
26–52	−61.5	−59.8
52–104	−46.0	−45.5
104–208	−30.3	−30.8
208–416	−13.6	−13.2
416–832	+0.3	+1.0

The noise power rises by 15.5, 15.7 and 16.7 dB per doubling of the band, in line with second-order shaping (about 15 dB per octave) and within 2 dB of the model in every band. Harmonics: HD2 −77.8 dBc (model −76.9), HD3 −72.0 dBc (−68.3), HD4 −65.7 (−67.3), HD5 −61.6 (−63.8); all lie outside the 25 kHz band at this input frequency (HD2 sits just beyond its edge, at bin 26), so they do not reduce the in-band SNDR figure. The strongest non-signal bin inside the band is −79.8 dBc, at the band edge.

7.3 Limits of this result

- The transient simulation contains *no noise sources*, and neither does the model; the SNDR is limited by quantization and deterministic non-idealities. kT/C, OTA thermal and comparator noise are not included.
- One amplitude, one input frequency, TT, 27 °C; no amplitude sweep, no PVT.
- Block5_SH is characterized standalone. The modulator uses the same gate cells inside its stage cores, so E1–E8 describe the gate but were not run inside the loop.
- The OTA of the modulator netlists is the `OTA_CMFB` variant, whose sizing differs from the OTA characterized in the Block 1 report; see the Revision 2 addendum of that report. The loop result is therefore not a verification of the Block 1 OTA.

8 Specification Decision

The behavioral model (noiseless, second order, 1 bit, median over 8 input phases, in-band signal at about half the band edge) gives:

Table 6: Model SNDR versus oversampling ratio at $f_s = 4$ MHz.

OSR	Bandwidth	SNDR at −6 dBFS	SNDR at −3 dBFS
80	25.0 kHz	77.0 dB	74.3 dB
128	15.6 kHz	84.7 dB	82.1 dB
160	12.5 kHz	90.1 dB	90.5 dB
256	7.8 kHz	95.5 dB	96.3 dB

The original 99-dB target is therefore not reachable at OSR ≤ 256 with this architecture. On 2 November 2026 the project owner set the specification to **≈75 dB SNDR at OSR 80,** 25 kHz **bandwidth** (about 12 bits), which matches the measured 75.4 dB. Higher OSR, a higher-order

loop or a multi-bit quantizer remain options for a future revision; thermal noise (kT/C) was not evaluated and may limit the higher-OSR options.

9 Decisions, Known Limits and Closure

Table 7: Decisions of the project owner.

Date	Subject	Decision
2 Nov 2026	Hold error	Pedestal accepted as measured, not final; no numeric limit set.
2 Nov 2026	OTA compensation check	Closed with the gap on the loop OTA recorded as a known limit.
2 Nov 2026	Specification	≈75 dB at OSR 80, 25 kHz.
2 Nov 2026	Block closure	Approved for the scope nominal, pre-layout, simulation.

Known limits (not verified).

- No transient noise; no amplitude sweep; no PVT, mismatch or post-layout evidence.
- Standalone track-and-hold results (E1–E8) were not repeated inside the loop.
- The original timing of the 450 ns anomaly is not available; the aperture explanation is a hypothesis.
- The OTA in the modulator netlists differs from the Block 1 OTA; its AC characterization with the switched-capacitor common-mode feedback closed, at its loop bias, is open. The Block 1 addendum shows that the compensation capacitance alone (MF 80 to 100) changes UGF by −14% and phase margin by +5.2° on the Block 1 OTA.
- DC offset was verified only through the 100 mV test (error 0.2 mV); no zero-input test.
- The transistor sizing of the integrators, comparators and common-mode feedback was not changed.

The next block (the non-overlapping clock generator, Block 6) starts only on top of this closed block; its first requirement is the 3 ns input-stability margin of Section 4.

A Reproducibility Notes

Testbenches: `tb_5F_E1_baseline` to `E8_level_sweep` (generated from the repository template by `tools/make_tb.py`; connectivity checked by `tools/check_sch.py`) and `tb_MIM_C_AC`. Analysis: `tools/run_tahap2_3.py` (subcommands `netlist`, `sim`, `analyze-hold`, `analyze-mim`, `analyze-aperture`, `analyze-level`, `selftest`). Closed loop: `run_closed_loop.py` with

```
-run -name sine300_n4096 -cycles 4112 -reset-us 5 -amp-mv 300 -bin 13 -osr 80 -model
```

which produces `metrics.json`, `manifest.json`, `bitstream.txt`, `ngspice.log` and `waveform.dat`. The number of cycles is chosen so that the analysed record is a power of two ($4112 - 16 = 4096$). All figures in this report were drawn from these data files. Numbers marked as estimates (the third-harmonic estimate of Section 5) are derived quantities, not simulation outputs.